

GUJARAT TECHNOLOGICAL UNIVERSITY
BE –SEMESTER 1&2(NEW SYLLABUS)EXAMINATION- WINTER 2018

Subject Code: 3110011**Date: 04-01-2019****Subject Name: Physics****Time: 10:30 am to 01:00 pm****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

	Marks
Q.1 (a) Define: Ductility and Plasticity.	03
(b) Draw: Stress – Strain diagram with necessary notation. Explain: Elastic Limit and Upper Yield Point in detail.	04
(c) Explain Types of Elasticity In detail. Explain Factor affecting on Elasticity.	07
Q.2 (a) Define :Damped Harmonic Motion	03
(b) Explain Sound Absorption and Reverberation	04
(c) Illustrate various aspect associated with Acoustic of building.	07
OR	
(c) Derive the Expression for Depression of Cantilever.	07
Q.3 (a) Write down various advantage and disadvantage of NDT.	03
(b) Calculate the frequency to which a piezo electrical oscillator circuit should be tuned so that a piezo electrical crystal of 0.1cm thickness vibrates in its fundamental mode to generate ultrasonic waves. (Young`s modulus and density of the materials of crystal are 80GPa and 2654 Kg/m ³)	04
(c) Write a short note on Piezoelectric method for production of ultrasonic sound.	07
OR	
Q.3 (a) Explain Hook`s Law	03
(b) Calculate the fundamental frequency of vibration when A quartz crystal of 0.15cm thickness is vibrating at resonance. (Young`s Modulus of quartz is 7.9×10^{10} N/m ² and density is 2650 Kg/m ³)	04
(c) Explain: Non Destructive Testing.	07
Q.4 (a) Define: Penetration depth in the vicinity of Superconductivity.	03
(b) The Critical temperature of superconductor is 9.15K. At zero Kelvin the critical field is 0.196 Tesla. Calculate the field at 6K.	04
(c) Explain Josephson`s Junction and its applications.	07
OR	
Q.4 (a) Define Ultrasonic waves with necessary properties.	03
(b) Explain various applications of superconductor.	04
(c) Write short note Ruby Laser with necessary diagram.	07
Q.5 (a) Define: Population Inversion	03

- (b) Differentiate normal light and Laser light. **04**
(c) Explain various applications of LASER in Engineering and Medicine **07**

OR

- Q.5** (a) Define : Simulated Emission **03**
(b) Explain Kundt's tubes method for the detection of Ultrasonic Sound. **04**
(c) The volume of room is 800m^3 . The wall area of the room is 240m^2 , the floor area is 120m^2 and the ceiling area is 120m^2 . The average sound absorption coefficient, (a) for the walls is 0.03 (b) for the ceiling is 0.8 (c) the floor it is 0.06 .Calculate the average sound absorption coefficient and the reverberation time. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-I & II (NEW) EXAMINATION – SUMMER-2019****Subject Code: 3110011****Date: 03/06/2019****Subject Name: Physics****Time: 10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

Marks

- Q.1**
- (a) An elastic rod having diameter of 30 mm, 10 cm long extends by 2.5 cm under tensile load of 28 kN. Find the stress, strain and the Young's modulus for the material of the rod. **03**
- (b) Match column I with column II and select the correct choice. **04**
- | I | | II | |
|---|--------------|----|--------------------|
| 1 | W. C. Sabine | A | Ruby Laser |
| 2 | T. Maiman | B | Elastic constant |
| 3 | H. K. Onnes | C | Reverberation time |
| 4 | Robert Hooke | D | Superconductors |
- (c) Explain in detail construction and working of Ruby Laser with the help of necessary schematic and energy level diagram. **07**
- Q.2**
- (a) An ultrasonic source of 150 KHz sends down a pulse towards the seabed, which returns after 0.82 s. The velocity of ultrasound in sea water is 1700 m/s. Calculate the depth of sea and wavelength of pulse. **03**
- (b) Differentiate Type – I and Type – II superconductors. (Any Four) **04**
- (c) Explain in detail Stress – Strain diagram for a bar or wire. **07**
- OR**
- (c) (1) Calculate the critical current for a superconducting wire of lead having diameter of 1 mm at 4.5 K. Critical temperature for lead is 7.2 K and critical magnetic field at 0 K is 6.5×10^4 A/m. **04**
- (2) What is Meissner effect? Prove that superconductors are perfect diamagnetic materials. **03**
- Q.3**
- (a) Find the acceleration of particle performing simple harmonic motion (SHM) when it is at 0.6 m from its mean position. The time period of SHM is 0.05 sec. Also calculate maximum velocity if the amplitude of SHM is 2 m. **03**
- (b) List various properties of superconductor. Explain in brief any three properties out of them. **04**
- (c) Define piezoelectric effect and explain in detail piezoelectric ultrasonic generator with necessary circuit diagram. Also mention its merit and demerit. **07**

OR

Q.3	(a)	Calculate length of an iron rod which can be used to produce ultrasound of frequency 22 kHz. Given that Young's Modulus and density of iron is $11.6 \times 10^{10} \text{ N/m}^2$ and $7.25 \times 10^3 \text{ kg/m}^3$, respectively.	03
	(b)	Differentiate Free and Forced oscillations. (Any Four)	04
	(c)	(1) Write short note on Reverberation and Reverberation time.	03
		(2) Define Magnetostriction effect and draw the neat and clean circuit diagram of Magnetostriction ultrasonic generator.	03
		(3) Classify the sound based on its frequencies.	01
Q.4	(a)	In the acoustic grating experiment, the wavelength of light transmitted through a liquid is 650 nm. The 1 st order angle of diffraction is 0.152° . Calculate the velocity of ultrasound in the liquid. The frequency of the ultrasound is 2.5 MHz.	03
	(b)	What is the full form of LASER? Give applications of Laser in various fields.	04
	(c)	(1) Differentiate Destructive and Non-destructive testing methods. (Any Four)	04
		(2) List few applications of superconductors and explain in brief SQUID.	03
OR			
Q.4	(a)	Discuss elastic behavior of solid materials.	03
	(b)	A solid disc of 1 kg mass and 0.2 m diameter is suspended in a horizontal plane by a vertical wire attached to its center. The length and diameter of the wire is 1.5 m and 2 mm, respectively. Calculate modulus of rigidity of wire and the time period Torsional oscillations if Torsional rigidity $C_s = 7.8 \times 10^{-3} \text{ kg}\cdot\text{m}^2/\text{s}^2$.	04
	(c)	What is Damped and Undamped vibrations? Derive the differential equation and general solution of damped harmonic motion.	07
Q.5	(a)	Calculate the wavelength of Laser light if the separation between metastable state and lower energy state is 1.80 eV. (consider Planck's constant is $4.14 \times 10^{-15} \text{ eV}\cdot\text{s}$)	03
	(b)	Derive the relation between Einstein's coefficients with necessary assumptions.	04
	(c)	(1) Define Following:	03
		(a) Poisson's ratio	
		(b) Weber-Fetchner law	
		(c) Population inversion	
		(2) List the fundamental components of the Laser and draw the block diagram of Laser consists them.	02
		(3) What is Ultrasound? List various methods of detecting ultrasonic waves.	02
OR			
Q.5	(a)	Discuss various factors affecting the elasticity.	03
	(b)	Explain in brief SONAR and its application.	04

- (c) (1) The volume of room is 800 m^3 . The wall area of the room is 230 m^2 , the floor and ceiling area is 130 m^2 . The absorption coefficient of wall, floor and ceiling is 0.05, 0.75, and 0.09, respectively. Calculate the total absorption, average absorption coefficient and the reverberation time **04**
- (2) Define following: **03**
- (a) Viscosity
 - (b) London penetration depth
 - (c) Metastable state

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER– I & II (NEW) EXAMINATION – WINTER 2019****Subject Code: 3110011****Date: 02/01/2020****Subject Name: Physics****Time: 10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

		Marks
Q.1	(a) Define: coefficient of viscosity.	03
	(b) Define Reverberation time and write down the Sabine's formula of it by explaining the parameters in it.	04
	(c) What is damping motion? Derive the differential equation and general solution of damped harmonic motion.	07
Q.2	(a) Define resonance in an oscillating system.	03
	(b) Draw stress- strain diagram. Explain the main points of it.	04
	(c) Write down the factors affecting the acoustics of an auditorium. Give remedies.	07
	OR	
	(c) Explain Young's Modulus, shear modulus, bulk modulus and Poisson's ratio.	07
Q.3	(a) Write down various applications of ultrasonic waves.	03
	(b) Calculate the natural frequency of 50 mm length of a pure iron rod. Given that Young's modulus of iron = 11.5×10^{10} N/m ² and density of pure iron = 7.25×10^3 kg/m ³ . Can you use it in a magnetostriction oscillator to produce ultrasonic waves?	04
	(c) Describe production of ultrasonic waves by magnetostriction method. Give its advantages and limitations.	07
	OR	
Q.3	(a) What do you understand by NDT. Give names of few NDT methods.	03
	(b) A Nickel rod having 5 cm length is vibrating at resonance. Calculate the fundamental frequency of vibration for which ultrasonic waves are generated. Given that Young's modulus of Nickel = 2.14×10^{11} N/m ² , density of Nickel = 8.908×10^3 kg/m ³ .	04
	(c) Describe acoustic diffraction method to determine the speed of sound in liquid with suitable diagram.	07
Q.4	(a) Explain the phenomenon of superconductivity.	03
	(b) The critical magnetic field of Niobium is 1×10^5 Tesla at 8 K and 2×10^5 Tesla at 0 K. Calculate the transition temperature of the element.	04
	(c) Write down the applications of superconductors.	07
	OR	
Q.4	(a) Describe BCS theory of superconductivity.	03
	(b) Derive the formula for time period of a torsional pendulum.	04
	(c) (i) Josephson Junction and its applications 04	07
	(ii) What force is required to stretch a steel wire to double the length when its area of cross section is 1 cm ² . Given that Young's modulus of wire is 7×10^{10} N/m ² .	03

- Q.5** (a) Write down the properties of LASER light. **03**
(b) What is the wavelength of light of Ruby Laser if the separation between metastable state and lower energy state is 1.79 eV. Given that Planck's constant = 6.64×10^{-34} Js. **04**
(c) Describe the construction and working principle of He-Ne LASER with suitable diagrams. **07**

OR

- Q.5** (a) Classify sound on the basis of frequency with suitable examples. **03**
(b) Derive the relationship between Einstein Coefficients. **04**
(c) Write down the various applications of LASER. **07**

Seat No.: _____

Enrolment No. _____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I & II (NEW) EXAMINATION – WINTER 2020

Subject Code:3110011

Date:17/03/2021

Subject Name:Physics

Time:10:30 AM TO 12:30 PM

Total Marks:56

Instructions:

1. Attempt any FOUR questions out of EIGHT questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

MARKS

- | | | | |
|------------|-----|---|----------|
| Q.1 | (a) | A load of 2.6 kg produces an extension of 1 mm in a wire of 3.082 m and 1 mm in diameter. Calculate the Young's modulus of a wire. | 3 |
| | (b) | Define and describe Elasticity and Plasticity in detail by giving suitable examples. | 4 |
| | (c) | Explain the terms: (i) Young's modulus, (ii) Bulk modulus, (iii) Shear modulus and (iv) poisson's ratio. Show any one relation between different moduli of elasticity. | 7 |
| Q.2 | (a) | An empty assembly hall have a volume of 11260 m ³ and its total absorption is equivalent to 92 m ² of open window. What will be the effect on reverberation time if an audience fills the assembly hall full and thereby increase the absorption by another 92 m ² of open window? | 3 |
| | (b) | Define and differentiate between transverse and longitudinal waves. | 4 |
| | (c) | Describe Simple Harmonic motion with some examples. Differentiate Free and Forced oscillations (any four points). | 7 |
| Q.3 | (a) | Describe any three applications of Ultrasonics. | 3 |
| | (b) | Write short note on (i) Ultrasonics and (ii) Non-Destructive Testing | 4 |
| | (c) | Write in detail Piezoelectric method for the production of ultrasonic sound using quartz crystal. Give suitable diagram and circuit for the same. | 7 |
| Q.4 | (a) | Superconducting Gadolinium has a critical temperature of 8.6 K at magnetic field of 12.95 Tesla. Find its critical field at 4.1 K. | 3 |
| | (b) | What are superconductors? Explain any three properties of superconductors. | 4 |
| | (c) | Explain in brief (i) Josephson's junction and its applications, (ii) SQUID | 7 |
| Q.5 | (a) | Describe and differentiate ordinary light and Laser light | 3 |
| | (b) | Write properties of Laser light. | 4 |
| | (c) | What is the full form of LASER? Explain in detail construction and working of Ruby Laser with the help of necessary schematic and energy level diagram. | 7 |

- Q.6** (a) Describe population inversion with suitable diagram. 3
- (b) State two advantage and disadvantages of non-destructive testing method 4
- (c) Draw: Stress – Strain diagram with necessary notation. Explain: Elastic Limit and Upper Yield Point in detail. 7
- Q.7** (a) Describe role of stimulated emission in lasing action. 3
- (b) Describe four methods for detection of ultrasonics. 4
- (c) State the acoustic requirements of a good auditorium. Write factors affecting architectural acoustics with remedies. 7
- Q.8** (a) Describe any three property of ultrasonic sound. 3
- (b) Describe ultrasonic flaw detector. How is it used in detection of flows in metals? 4
- (c) What is Cantilever? Obtain the expression for depression at free end of thin beam clamped horizontally at one end and loaded at other end. 7

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-1/2 EXAMINATION – WINTER 2021****Subject Code:3110011****Date:22/03/2022****Subject Name:Physics****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Define Hooke's law, stress and strain.	03
	(b) Explain various properties of LASER beam.	04
	(c) Describe Stress-Strain diagram in detail.	07
Q.2	(a) Define wave motion. Discuss different types of waves.	03
	(b) (i) A cinema hall has a volume of $9,500 \text{ m}^3$. What should be the total absorption in the hall if the reverberation time of 1.7 s is to be maintained?	02
	(ii) An ultrasonic source of 0.075 MHz sends down a pulse towards the seabed, which returns after 0.95 s. The velocity of ultrasound in sea water is 1800 m/s. Calculate the depth of the sea and wavelength of pulse.	02
	(c) Discuss in detail the different types of elasticity. List different factors affecting elasticity.	07
	OR	
	(c) What are the factors affecting acoustics of the building and give their remedies.	07
Q.3	(a) A brass bar having a cross-section of 1 cm^2 is supported on two knife-edges 1.5 m apart. A load of 2 kg at the center of the bar depresses that point by 2.75 mm. What is the Young's Modulus for brass?	03
	(b) Describe viscosity. How the comparison of viscosities of two liquids can be done?	04
	(c) Establish the relation between Einstein's coefficients.	07
	OR	
Q.3	(a) In the acoustic grating experiment, the wavelength of light transmitted through a liquid is $5970 \text{ \AA}$. The 1 st order angle of diffraction is 0.195° . Calculate the velocity of ultrasound in the liquid having frequency 2.7 MHz.	03
	(b) Expand SQUID. How is it formed and give it's applications.	04
	(c) Explain in detail the production of ultrasonic waves through piezoelectric oscillator method.	07
Q.4	(a) Calculate the frequency to which piezoelectric oscillator circuit should be tuned so that a piezoelectric crystal of thickness 0.2 cm vibrates in it's fundamental mode to generate ultrasonic waves. Young's modulus is 80 Gpa and density of material is 2654 kg/m^3	03
	(b) Define Cooper pair. Explain BCS theory for superconductors.	04
	(c) What are the applications of Ultrasound? Discuss them in detail.	07
	OR	

- Q.4** (a) The critical current density equal to $1.71 \times 10^8 \text{ A/m}^2$ is required to change a superconducting wire of radius 0.5 mm at 4.18 K. If the critical temperature of the material is 7.5 K, calculate the maximum value of the critical magnetic field. **03**
- (b) Explain NDT with it's objectives. **04**
- (c) Explain in detail construction and working of He-Ne Gas LASER with necessary schematic and energy level diagrams. **07**
- Q.5** (a) A voltage of $6.7 \mu\text{V}$ is applied across a Josephson junction. What is the frequency of the radiation emitted by the junction in GHz? **03**
- Planck's constant = $6.626 \times 10^{-34} \text{ J.s}$
- (b) Define Optical resonator, life time, metastable state and pumping mechanism for LASER. **04**
- (c) Define Superconductor. Discuss it's properties in detail. **07**
- OR**
- Q.5** (a) A hall has a volume of 2790 m^3 . It's total absorption is equivalent to 98.80 m^2 of open window. What will be the effect on reverberation time if the audience fill the hall and thereby increase the absorption by another 98.80 m^2 of open window. **03**
- (b) What is ultrasonic waves? Give properties and detection methods for ultrasonics. **04**
- (c) Give applications of LASER in various fields in detail. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II(NEW)EXAMINATION – SUMMER 2022****Subject Code:3110011****Date:04-08-2022****Subject Name:Physics****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Define ultrasonic waves. Mention two applications of Ultrasonic waves.	03
	(b) An oscillator is performing SHM with an acceleration of 0.2 m/s^2 when displacement is 5 cm. Its maximum speed is 0.15 m/s. Find time period and amplitude of oscillator.	04
	(c) Write applications of LASER in various field.	07
Q.2	(a) A steel wire has 3 m length and 0.5 mm radius. When it is stretched by force of 49 N find (i) longitudinal stress, (ii) longitudinal strain and (iii) elongation produced in the wire if Young's modulus of steel is $2.1 \times 10^{11} \text{ N/m}^2$. Take $g = 9.8 \text{ m/s}^2$	03
	(b) (i) Calculate the twisting couple of a solid shaft of 2 m length and 0.2 m diameter when it is twisted through an angle of 0.01 rad. The coefficient of rigidity of the material is $8 \times 10^{10} \text{ N/m}^2$. (ii) Calculate the twist produced in a solid wire of 5 m length and 0.1 m radius. The twisting couple produced in the wire is $2 \times 10^4 \text{ N m}$ and the coefficient of rigidity of the material is 10^{11} N/m^2 .	04
	(c) Draw stress versus strain graph with appropriate notations. Explain elastic limit and upper yield point in detail.	07
OR		
	(c) Describe the Ostwald's viscometer and explain how it can be used for comparison of viscosities of two liquids.	07
Q.3	(a) (i) Define Poisson's ratio. Write its expression.	02
	(ii) Define simple harmonic motion (SHM).	01
	(b) What is Damped Harmonic Motion? Derive differential equation for it.	04
	(c) Mention five factors affecting acoustic of building. Explain each and also mention their remedies.	07
OR		
Q.3	(a) A hall, having volume of 6000 m^3 , is found to have a reverberation time of 2.2 s. If the area of sound absorbing surface is 500 m^2 , calculate the absorption coefficient.	03

- (b) The volume of a room is 1500 m^3 . The wall area of the room is 240 m^2 , the floor area is 130 m^2 , and the ceiling area is 130 m^2 . The average sound absorption coefficient (i) for wall is 0.035; (ii) for the ceiling is 0.75; and (iii) the floor is 0.05. Calculate the average sound absorption coefficient and reverberation time. **04**
- (c) Describe, with appropriate diagram, principle, construction and working of Magnetostriction Generator (Oscillator). **07**
- Q.4 (a)** Explain the following: **03**
- (i) Population inversion
(ii) Metastable State
(iii) Active medium
- (b) An ultrasonic source of 0.09 MHz sends down a pulse towards the seabed which returns after 0.55 s. The velocity of sound in water is 1800 m/s. Calculate (i) the depth of the sea and (ii) wavelength of pulse. **04**
- (c) What is NDT? Write down 3 advantages and 3 disadvantages of NDT. **07**
- OR**
- Q.4 (a)** What is Meissner effect? For superconductor show that $\chi_m = -1$. **03**
- (b) Discuss the method to determine depth of the sea with the help of SONAR **04**
- (c) What is Josephson Effect? Explain DC Josephson effect and AC Josephson effect with necessary equations. **07**
- Q.5 (a)** For Hg of mass number 202, the T_C value is 4.2 K. Find the T_C value for the isotope of mercury of mass number 199.5. Take, $\alpha = 0.5$. **03**
- (b) The pulse arrival times from the steel bar of 30 cm thickness during the detection of possible defects using pulse echo method are 30 μs and 60 μs . Find out the distance of defect in steel bar from the entrance of ultrasonic wave. **04**
- (c) (i) Calculate the penetration depth of lead at 6.2 K if the London penetration depth at 0 K is 36.5 nm. The critical temperature of lead is 7.2 K. **07**
- (ii) Critical magnetic field of Lead is $6.5 \times 10^3 \text{ A/m}$ at 0 K. Calculate the temperature at which critical magnetic field of Lead drop to $3 \times 10^3 \text{ A/m}$. The critical temperature of Lead is 7.2 K. Calculate critical current density at that temperature if radius of the wire is 0.5 mm.
- OR**
- Q.5 (a)** Write and explain any three properties of LASER. **03**
- (b) Compare spontaneous emission and stimulated emission. (Mention four points of comparison) **04**
- (c) Explain construction and working of Ruby LASER. **07**
- *****

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-I & II(NEW) EXAMINATION – WINTER 2022****Subject Code:3110011****Date:04-03-2023****Subject Name:Physics****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Discuss stress-strain diagram.	03
	(b) A wire of length 2 m and radius 0.5 mm is clamped at one of its ends. Calculate the torque required to twist the other end by 90°. The rigidity modulus of the wire is $2.8 \times 10^{10} \text{ N/m}^2$.	04
	(c) Derive an expression for the depression of the free end of cantilever. What are the applications of cantilevers?	07
Q.2	(a) Define the term: (1) Reverberation time (2) Piezoelectric effect (3) Acoustic grating.	03
	(b) A hall has a volume of 2265.6 m^3 . Its total absorption is equivalent to 92.92 OWU. What will be the effect on the reverberation time if curtains of total absorption equivalent to 92.90 OWU are placed inside?	04
	(c) What is simple harmonic motion? Obtain equation for kinetic energy and potential energy of simple harmonic oscillator. Show that at any time the sum of kinetic energy and potential energy remains constant.	07
	OR	
	(c) Derive differential equation for forced oscillator. Explain transient behavior and steady state behavior with necessary equation.	07
Q.3	(a) A piezoelectric X-cut crystal plate has a thickness of 1.6 mm. If the velocity of propagation of sound waves along the X-direction is 5760 m/s, calculate the fundamental frequency of the crystal.	03
	(b) Explain the process of non-destructive testing of materials using ultrasonic waves by pulse echo system.	04
	(c) What is meant by magnetostriction effect? Explain how the ultrasonic waves are produced using magnetostriction oscillator.	07
	OR	
Q.3	(a) Write the properties of ultrasound wave.	03
	(b) Calculate the value of capacitance required to produce ultrasonic waves of 10^8 Hz with an inductance of 1 henry.	04
	(c) Described the different methods used for detection of ultrasonic waves.	07
Q.4	(a) How is superconductivity affected by magnetic field and current?	03
	(b) Distinguish between Type - I and Type - II superconductor.	05
	(c) What is BCS theory? Enumerate the important results of BCS theory.	06
	OR	
Q.4	(a) Explain how cooper pair state is responsible for superconductivity?	03
	(b) The magnetic field intensity in the tin material is $\frac{3 \times 10^5}{4\pi} \text{ A/m}$ at 0 K. Calculate the temperature of the superconductor if the field intensity is measured	04

as $\frac{2 \times 10^5}{4\pi}$ A/m. Critical temperature of tin is 3.69 K.

(c) Explain the properties of superconducting materials in detail. **07**

Q.5 (a) Write the advantages of using laser drilling in industries. **03**

(b) Using Einstein's theory, show that in the optical region, say, at $\lambda=5000 \text{ \AA}$ and $T=300 \text{ K}$, the amplification is not possible. **04**

(c) Explain with neat sketch the construction and working of ruby laser. **07**

OR

Q.5 (a) In Ruby laser, total number of Cr^{+3} ions in excited state are 2.8×10^{19} . If the laser emits radiation of wavelength $7000 \text{ \AA}$, calculate energy of laser pulse. **03**

(b) What is population inversion? Explain the different methods for achieving population. **04**

(c) What is a gas laser? Explain the working of He-Ne laser with relevant diagrams. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II(NEW) EXAMINATION – SUMMER 2023****Subject Code:3110011****Date:01-08-2023****Subject Name:Physics****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Define Ductility, Brittleness and Elasticity.	03
	(b) Define Spontaneous and Stimulated emission.	04
	(c) Derive an expression for Twisting couple in a wire.	07
Q.2	(a) Define simple harmonic motion (SHM). Give Any two examples of it.	03
	(b) Find the acceleration of particle performing simple harmonic motion when it is at 0.6 m from its mean position. The time period of S.H.M. is 0.05 sec. also calculate maximum velocity if the amplitude of S.H.M. is 2 m.	04
	(c) Derive an expression for depression of cantilever.	07
	OR	
	(c) Explain classification and properties of sound absorbing material.	07
Q.3	(a) Calculate Poisson's ratio and the rigidity modulus of copper using following data : Young's modulus $10.5 \times 10^{10} \text{ Nm}^{-2}$ and the bulk modulus of copper $14.3 \times 10^{10} \text{ Nm}^{-2}$.	03
	(b) Write a short note on I shaped girders.	04
	(c) Explain construction and working of He – Ne laser.	07
	OR	
Q.3	(a) An ultrasonic source of 0.09 MHz sends down a Pulse towards the seabed which returns after 0.55 sec. The velocity of sound in water is 1800 m/s. Calculate the depth of the sea and wavelength of pulse.	03
	(b) Compare Type –I and Type – II superconductors. (Any four points).	04
	(c) Explain in detail Magnetostriction method for production of ultrasonic waves.	07
Q.4	(a) Find the frequency of the first and second modes of vibration for a quartz crystal of piezoelectric oscillator. The velocity of longitudinal waves in quartz is $5.5 \times 10^3 \text{ m/s}$. thickness of quartz crystal is 0.05 m.	03
	(b) Explain Josephson junction.	04
	(c) Discuss Ultrasonic flaw detector method for NDT, with its advantages and limitations.	07
	OR	
Q.4	(a) The critical temperature of mercury with isotopic mass 199.5 is 4.185 K. calculate its critical temperature when its isotopic mass is 203.4.	03
	(b) Compare Non destructive test (NDT) with destructive test (DT). (any four points)	04

- (c) Explain basic components of laser generation with diagram. Also give types of laser. **07**
- Q.5** (a) Calculate the critical current for a wire of Pb having a diameter of 1 mm at 4.5 K. The critical temperature for Pb is 7.2 K and $H_c(0) = H(0) = 6.5 \times 10^4$ A/m. **03**
- (b) Define pumping with its types. **04**
- (c) Explain Meissner effect. Hence prove that superconductors are perfect diamagnetic materials. **07**
- OR**
- Q.5** (a) A hall has a volume of $1,20,000 \text{ m}^3$, it has reverberation time of 1.5 sec. what is average absorbing power of the surface if the total absorbing surface area is $25,000 \text{ m}^2$. **03**
- (b) Explain working of SONAR. **04**
- (c) Explain construction and working of Ruby laser with necessary diagrams. **07**

Seat No.: _____

Enrolment No. _____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER-I (NEW) EXAMINATION – WINTER 2023

Subject Code:3110011

Date:12-01-2024

Subject Name:Physics

Time:02:30 PM TO 05:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

Marks

- Q.1**
- | | | |
|-----|--|-----------|
| (a) | Explain the term bulk modulus, modulus of rigidity and yield point. | 03 |
| (b) | Derive the formula for time period of a torsional pendulum. | 04 |
| (c) | What is Cantilever? Obtain the expression for depression at free end of thin beam clamped horizontally at one end and loaded at other end. | 07 |

- Q.2**
- | | | |
|-----|--|-----------|
| (a) | Explain sound absorption and reverberation. | 03 |
| (b) | Discuss in detail the forced vibration and amplitude resonance. | 04 |
| (c) | What is damping motion? Derive the differential equation and general solution of damped harmonic motion. | 07 |

OR

- | | | |
|-----|---|-----------|
| (c) | Discuss in detail the different types of elasticity. List different factors affecting elasticity. | 07 |
|-----|---|-----------|

- Q.3**
- | | | |
|-----|--|-----------|
| (a) | Describe any three property of ultrasonic sound. | 03 |
| (b) | Calculate the length of nickel rod required to produce ultrasonic waves of frequencies of 50 kHz and 100 kHz. Consider the fundamental mode of vibrations of the rod. Given, density of nickel= 8908 kg/m ³ , Young's modulus of the nickel= 2.14×10^{11} N/m ² . | 04 |
| (c) | Describe production of ultrasonic waves by magnetostriction method. Give its advantages and limitations. | 07 |

OR

- Q.3**
- | | | |
|-----|---|-----------|
| (a) | An elastic rod having diameter of 30 mm, 10 cm long extends by 2.5 cm under tensile load of 28 kN. Find the stress, strain and the Young's modulus for the material of the rod. | 03 |
| (b) | A nickel rod having 5 cm length is vibrating at resonance. Calculate the fundamental frequency of vibration for which ultrasonic waves are generated. Given that Young's modulus of the nickel= 2.14×10^{11} N/m ² , density of nickel= 8.908×10^3 kg/m ³ . | 04 |

- (c) Explain in detail about the methods for the detection of ultrasound. Give any three applications of ultrasonic waves. **07**
- Q.4** (a) Explain the Josephson's junction and state any one of its application. **03**
 (b) Determine the critical current for a superconducting ring of diameter 10^{-3} m at temperature 4.2 K. Given the critical temperature (T_c) for sample is 7.18 K and critical magnetic field at 0 K ($H_c(0)$) is 6.5×10^4 A/m. **04**
 (c) Explain London penetration depth. Discuss various applications of superconductors. **07**
- OR**
- Q.4** (a) Explain NDT. Discuss advantages of NDT. **03**
 (b) A critical magnetic field of lead wire is 6.5×10^3 A/m at 0 K. At what temperature would the critical magnetic field of lead drop to 4.5×10^3 A/m if the critical temperature of lead is 7.18 K? What is the critical current density at that temperature if the diameter of the wire is 2×10^{-3} m? **04**
 (c) Explain in detail construction and working of Ruby LASER with necessary schematic and energy level diagrams. **07**
- Q.5** (a) In He-Ne laser system, the two energy levels of Ne involved in lasing action have energy values of 20.66 eV and 18.70 eV. Population inversion occurs between these two levels. What will be the wavelength of a laser beam produced? What will be the population of the metastable energy level with respect to the upper excited level at 27 °C? **03**
 (b) Discuss the properties of LASER. **04**
 (c) Describe the construction and working principle of He-Ne LASER with suitable diagrams. **07**
- OR**
- Q.5** (a) Discuss any three important applications of LASER. **03**
 (b) Explain the term (i) stimulated emission, (ii) population inversion, (iii) pumping and (iv) metastable state. **04**
 (c) (i) What will be the resultant sound level when 70 dB sound is added to a 80 dB sound? **04**
 (ii) A particle execute simple harmonic motion with amplitude of 0.1 m and a period of 0.5 s. Calculate its displacement, velocity and acceleration (1/12) s after it crosses the mean position. **03**

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER-I & II (NEW) EXAMINATION – SUMMER 2024

Subject Code:3110011

Date:10-07-2024

Subject Name:Physics

Time:02:30 PM TO 05:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

- Q.1** (a) What is Meissner effect? For superconductor show that $\chi_m = -1$. **03**
(b) Draw Stress versus Strain diagram with necessary notation and discuss it. **04**
(c) (i) Write principle of Piezoelectric Generator. **07**
(ii) With proper circuit diagram describe construction and working of Piezoelectric Generator.
(iii) List merits and demerits of Piezoelectric Generator.
- Q.2** (a) What is reverberation and reverberation time? Write Sabine's formula for reverberation time and explain each term. **03**
(b) State any four points of differences between spontaneous emission and stimulated emission. **04**
(c) What is torsional pendulum? Derive expression of time period for vibration of torsional pendulum. **07**
- OR
- (c) Derive the expression for depression of Cantilever loaded at free end. **07**
- Q.3** (a) Write any three properties of ultrasonic waves. **03**
(b) Calculate the length of iron rod which can be used to produce ultrasonic waves of 75 kHz. Given that the density of iron is 7230 kg/m³ and its Young's modulus is 6×10^9 N/m². **04**
(c) What is called Acoustic Grating? Discuss the method to determine velocity of ultrasonic waves with the help of acoustic grating. **07**
- OR
- Q.3** (a) Give classification of sound waves based on frequency of wave. **03**
(b) Write any four points of differences between Destructive test and Non-destructive test. **04**
(c) (i) Discuss the method to determine depth of the sea with the help of SONAR. **07**
(ii) An ultrasonic wave of 7.2 MHz sends down a pulse towards the seabed, which returns after 1.2 s. The velocity of sound in seawater is 1800 m/s. Calculate the depth of sea and wavelength of pulse.

- Q.4** (a) Write any three applications of LASER. **03**
 (b) Write any four factors affecting acoustic of building and their remedies. **04**
 (c) Write detailed note on Ruby laser with proper diagram. **07**
- OR
- Q.4** (a) Write full form of LASER and discuss in brief characteristics of laser light. **03**
 (b) The volume of a room is 1500 m^3 . The wall area of the room is 240 m^2 , the floor area is 125 m^2 , and the ceiling area is 125 m^2 . The average sound absorption coefficient (i) for wall is 0.025; (ii) for the ceiling is 0.85; and (iii) the floor is 0.09. Calculate the average sound absorption coefficient and reverberation time. **04**
 (c) Write detailed note on He-Ne laser with proper diagram. **07**
- Q.5** (a) For a superconducting material isotopic mass is 195 amu and critical temperature is 4.5 K. Calculate isotopic mass at 5.7 K. **03**
 (b) Write any four points of comparison of type-I and type-II superconductor. **04**
 (c) What is superconductivity? Discuss any six properties of superconductor. **07**
- OR
- Q.5** (a) The critical current passing through a superconducting wire of radius 5 mm is 21 A. Calculate critical magnetic field. **03**
 (b) Explain in detail BCS theory of superconductivity. **04**
 (c) (i) Write a short note on SQUID. **07**
 (ii) Write short note on London Penetration Depth.

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II EXAMINATION – WINTER 2024****Subject Code:3110011****Date:04-01-2025****Subject Name:Physics****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	Marks
Q.1 (a) Calculate the twisting couple on a solid shaft of length 1.5 m and diameter 120 mm when it is twisted through an angle 3° . The coefficient of rigidity for the material of the shaft may be taken to be $93 \times 10^9 \text{ N/m}^2$.	03
(b) (i) What is Young's modulus? Give its equation. (ii) What is Bulk modulus? Give its equation. (iii) What is modulus of rigidity? Give its equation. (iv) What is Poisson's ratio? Give its equation.	04
(c) Explain the Stress-Strain diagram in detail.	07
Q.2 (a) Define sound absorption coefficient. Define reverberation time.	03
(b) The volume of room is 1500 m^3 . The wall area of the room is 260 m^2 , the floor area is 140 m^2 and the ceiling area is 140 m^2 . The average sound absorption coefficient for wall is 0.03, for the ceiling is 0.8 and for the floor is 0.06. Calculate the average absorption coefficient and the reverberation time.	04
(c) Explain in detail about characteristics of SHM.	07
OR	
(c) Discuss various factors affecting the acoustics of buildings and give their remedies.	07
Q.3 (a) Distinguish between transverse and longitudinal wave?	03
(b) What are ultrasonic waves? List out the properties of ultrasound.	04
(c) Explain with required circuit diagram the generation of ultrasonic waves with piezoelectric effect.	07
OR	
Q.3 (a) Calculate the length of an iron rod which can be used to produce ultrasonic waves of 20 KHz. Density of iron is $7.23 \times 10^3 \text{ kg/m}^3$ and Young's modulus is $11.6 \times 10^{10} \text{ N/m}^2$.	03
(b) Briefly discuss the applications of ultrasound.	04
(c) Explain in detail the method for measuring velocity of ultrasound using acoustical grating method.	07
Q.4 (a) The critical temperature of Nb is 9.15 K. At zero kelvin the critical field is 0.196 tesla. Calculate the critical field at 6K.	03
(b) Compare Type-I and Type-II superconductor.	04

- (c) What is meant by superconductivity? Describe the effect of (1) magnetic field, (2) isotope, (3) impurity, (4) current and (5) pressure on superconductors. **07**

OR

- Q.4** (a) Calculate the critical current which can flow through a long thin superconducting wire of aluminum of 10^{-3} m diameter. The critical magnetic field for aluminum is $7.9 \times 10^3 \text{ A}\cdot\text{m}^{-1}$. **03**

(b) Explain properties of Laser light. **04**

(c) Explain the construction and working of ruby laser. **07**

- Q.5** (a) A typical laser emits a $6940 \text{ \AA}$ wavelength. Find the energy difference (in eV) between the energy levels of this laser. **03**

(b) Explain various applications of LASER. **04**

(c) List out various advantage and disadvantage of NDT. **07**

OR

- Q.5** (a) What is cantilever? Write expressions for depression of cantilever when the load is fixed at the center for rectangular and circular bar. **03**

(b) Explain population inversion with suitable diagram. **04**

(c) Give a brief account of few important applications of superconductors. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I&II EXAMINATION – SUMMER 2025****Subject Code:3110011****Date:25-06-2025****Subject Name:Physics****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Define: Stress, Strain, and Spring constant.	03
	(b) A hall has a volume of 2265 m^3 . Its total absorption is equivalent to $94.85 \text{ sabine}\cdot\text{m}^2$. What will be the effect on reverberation time if the audience fill the hall, and thereby increasing the absorption by another $94.85 \text{ sabine}\cdot\text{m}^2$?	04
	(c) What is superconductor? Discuss about the Meissner effect, the effect of magnetic fields and the Isotopic effect in superconductors.	07
Q.2	(a) State any six properties of ultrasound.	03
	(b) Calculate the critical current for a superconducting wire made of lead, having a diameter of 1 mm at 4.2 K. Critical temperature (T_C) for lead is 7.18 K and $H_0 = 6.5 \times 10^4 \text{ A/m}$.	04
	(c) Discuss the construction and working of He-Ne laser.	07
	OR	
	(c) Discuss the construction and working of Ruby laser.	07
Q.3	(a) Find the force required to increase the length of a steel wire of 10^{-6} m^2 area of cross-section by 50% whose Young's modulus is $2 \times 10^{11} \text{ N/m}^2$.	03
	(b) Discuss the stress-strain curve for solid in detail.	04
	(c) Derive the differential equation for damped harmonic motion. Also, derive the solution for the case where $b^2 < \omega_0^2$.	07
	OR	
Q.3	(a) Derive the differential equation for simple harmonic motion.	03
	(b) Discuss any four factors affecting acoustics of a building and their remedies.	04
	(c) What is torsional pendulum? Derive the expression for the time period of vibration for torsional pendulum.	07
Q.4	(a) Explain the working of SONAR in detail.	03
	(b) Calculate the length of nickel rod required to produce ultrasonic waves with frequencies of 50 kHz and 100 kHz. Consider the fundamental mode of vibration for the rod. Given: Density of nickel = 8908 kg/m^3 , Young's modulus of nickel = $2.14 \times 10^{11} \text{ N/m}^2$.	04
	(c) Discuss the BCS theory to explain superconductivity in materials.	07
	OR	
Q.4	(a) State any six applications of superconductors.	03
	(b) Explain the construction and working of SQUID using Josephson junctions.	04
	(c) Discuss the construction and working of piezoelectric oscillator for the production of ultrasonic waves.	07

- | | | | |
|------------|------------|--|-----------|
| Q.5 | (a) | Define: Population inversion, Meta-stable state, and Pumping. | 03 |
| | (b) | State any eight applications of lasers in various fields. | 04 |
| | (c) | What is Non-destructive testing (NDT)? Discuss the use of ultrasound in NDT. | 07 |
| | | OR | |
| Q.5 | (a) | Define: Simple Harmonic Motion, Forced Vibration, and Free Vibration. | 03 |
| | (b) | Discuss about Young's modulus and bulk modulus in detail. | 04 |
| | (c) | What is Einstein's coefficient? Derive the relation between Einstein's coefficients. | 07 |

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II EXAMINATION – WINTER 2025****Subject Code:3110011****Date:16-01-2026****Subject Name:Physics****Time:02:30 PM TO 05:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) The critical temperature for lead is 7.1 K. At 6 K the superconducting property disappears if it is subjected to a magnetic field of 4.67×10^4 A/m. Find H_0 .	03
	(b) Explain properties Meissner Effect and Effect of magnetic field for Superconductors.	04
	(c) Define superconductivity. Describe its use in Maglev and Josephson Junction.	07
Q.2	(a) A hall of volume 1000 m^3 is found to have a reverberation time of 2.5 s. If the area of the sound absorbing surface is 5000 cm^2 , calculate the average absorption coefficient.	03
	(b) What is a wave? Describe Longitudinal and Transverse wave.	04
	(c) What is simple harmonic motion? Obtain its differential equation with graphical representation. Write equation for its potential energy and kinetic energy in terms of amplitude of a wave.	07
	OR	
	(c) What is damped harmonic motion. Solving its differential equation, obtain equation for its overdamped and underdamped conditions.	07
Q.3	(a) Write equation of bending of beam and explain I shaped girders with advantages.	03
	(b) A rectangular solid has dimensions $6 \times 6 \times 2 \text{ cm}$. A force of 0.2 N, applied tangentially to the upper surface, causes a displacement of 5 mm relative to the lower surface. Calculate the shearing strain, stress and rigidity modulus.	04
	(c) State types of moduli of elasticity. Derive relation of Young's modulus and modulus of rigidity.	07
	OR	
Q.3	(a) Define stress and strain. Give their equation and unit.	03
	(b) Calculate the Young's modulus in the cantilever depression method used. The length of the cantilever is 1 m which is suspended with a load of 150 gm. The depression is found to be 4 cm. The thickness of the beam is 5 mm and breadth of the beam is 3 cm.	04
	(c) What is twisting moment? Derive equation for twisting couple on a wire or shaft.	07
Q.4	(a) What is the ratio of the stimulated emission to spontaneous emission at a temperature of 250°C for the sodium D line?	03
	(b) Discuss properties of laser.	04
	(c) Obtain relation between Einstein's coefficients.	07

OR

- Q.4** (a) Write applications of Ultrasonic waves in various fields **03**
(b) List the properties of ultrasound. **04**
(c) Discuss with suitable diagram the piezoelectric method for ultrasonic wave generation. State its advantages and disadvantages. **07**
- Q.5** (a) Find the maximum velocity and acceleration of a particle performing SHM of period 20π s and amplitude 10 cm. **03**
(b) Discuss acoustic grating method for finding velocity of ultrasonic waves in medium. **04**
(c) What is destructive and non-destructive testing. Give difference between them. **07**
- OR**
- Q.5** (a) An Ultrasonic source of 0.07 MHz sends down a pulse towards the seabed which returns after 0.35 sec. The velocity of sound in water is 1800 m/s. Calculate the depth of sea and wavelength of pulse. **03**
(b) Write note on applications of laser in engineering and medicine. **04**
(c) Discuss Ruby laser for laser beam production with suitable diagram. **07**
